SUPPLEMENTARY MATERIAL

Deposition and Characterization of PVD Thin Films

4th Year Integrated MSc. Open Lab Report

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Procedure

Operation of the PVD System

1. The PVD system was initially kept in the **close position** (all the knobs) before starting the operation.
2. The **chiller** was switched on by first turning on the main supply and subsequently switching on the compressor.
3. The **rotary pump** was switched on using the rotary-pump control knob to establish the initial vacuum in the chamber, just after shifting the lever to the backing position.
4. The backing valve/lever was kept in the appropriate **backing position** during the initial pumping stage.
5. The rotary pump was allowed to operate until the required backing pressure was achieved. In the present procedure, the backing pressure was allowed to reach approximately

$$P_{\text{backing}} \approx 10^{-2}\ \text{mbar}. \tag{1}$$

6. After achieving the required backing pressure, the lever was shifted to the roughing position, and both the backing and roughing pressures were checked to be of the order of 10^{-2} mbar.
7. After achieving the required pressure on the Pirani gauge, the **diffusion pump** was started by keeping the lever in the backing position.
8. The system was allowed to operate for approximately **45 minutes** so that the diffusion pump could reach its proper operating condition.
9. The **high-vacuum valve** was then opened gradually to establish high vacuum in the deposition chamber.
10. The chamber pressure was allowed to decrease until a high-vacuum condition of approximately

$$P_{\text{chamber}} < 5 \times 10^{-5}\ \text{mbar} \tag{2}$$

was reached, as read on the Pirani gauge.

11. After obtaining the required vacuum, the substrate was positioned appropriately inside the deposition chamber by admitting air into the deposition chamber, before closing the high-vacuum valve and setting the Pirani gauge lever to the backing position.
12. The material to be deposited was loaded into the evaporation/deposition source (boat), after first being weighed.
13. After closing the chamber door, it was ensured that the air-admittance knob was closed; the high-vacuum valve was then reopened, with the Pirani gauge lever kept in the backing position.
14. The LT (low-tension) current was switched on and gradually increased until proper vaporisation of the source material was observed through the viewer glass.
15. After allowing some time for cooling, the high-vacuum valve was closed, the air-admittance valve was opened, and the chamber was then opened.
16. The deposited samples were removed from the chamber after completion of the deposition process and were retained for subsequent characterization.

Shutdown Procedure

17. After completion of the deposition, the **high-vacuum valve** was closed.
18. The diffusion pump was switched off using its corresponding control.
19. The system was returned to the **backing position** to maintain the appropriate pumping configuration during cooling.
20. The system was allowed to cool for approximately **30 minutes** before proceeding with further shutdown operations.
21. The rotary pump was then switched off.
22. The chiller was switched off after ensuring that the system had undergone the required cooling period.
23. Finally, the **main power supply** was switched off.

Work Done Till

Initial Deposition Trial

1. An initial deposition trial was carried out using a **glass substrate** and **Aluminum** as the target material.
2. We first cleaned the glass substrate using an **ultrasonic cleaner** with distilled water (10 mins.) and isopropanol (5 mins.) to remove surface contaminants and prepare the substrate for deposition.
3. Aluminum was then deposited onto the glass substrate, as shown in Fig. 1 at **35A** LT current, **55°C** chamber temperature and took ~**480 sec** in depositing.
4. This deposition process was done to provide us with an initial practical understanding of the operation of the PVD system and the formation of a thin film on a substrate.

Deposition on Silicon Substrate

1. Following the initial deposition on glass substrate, we selected a **p-type silicon substrate** ~(1*1, cm)(measured by software image and calc.) for the next deposition.We again cleaned the substrate using an **ultrasonic cleaner**.

2. **Si** (0.473g target taken) was subsequently deposited on the p-type Si substrate, as shown in Fig. 2 at **55A** LT current, **112°C** chamber temperature and took ~**550 sec** in depositing.

3. After deposition, two of the substrate pieces we attached showed a detectable and reasonable deposition of **Si**. However, the chamber was also exposed to Si, possibly due to the melting of the **tungsten holder**, which subsequently led to **improper vacuum formation** during the backing process.

4. This has now been fixed; the issue was traced to **oil present in the Pirani gauge**.

Plan for the Next Few Weeks

1. Following these trials, thin-film depositions will be carried out on a p-type Si substrate, comprising **two sample of (Si/Ge) and four sample of (Si/Ge/Si) layered**.

2. Annealing of the deposited samples will be carried out.

3. The **thickness** of the deposited films will be determined using the available characterization technique (step profiling).

4. **If time permits**, the **surface morphology** of the deposited films will be investigated using an appropriate microscopy technique, such as **optical microscopy, AFM, or SEM**, depending on availability.

5. The surface morphology will be analyzed in terms of film uniformity, grain structure, roughness, and other relevant features.

Summary of Progress

1. The basic operation of the PVD system has been learned.

2. The complete pumping sequence, including **backing-pump operation, diffusion-pump operation, high-vacuum operation, and system shutdown**, has been practiced.

3. An initial aluminum deposition was successfully carried out on a **glass substrate**.

4. Si was subsequently deposited on a **p-type silicon substrate**.

5. The deposited samples are now ready for systematic characterization.

6. The next stage of the work will focus on determining the **thickness of the deposited films** and **if time permits then surface morphology**.

Images

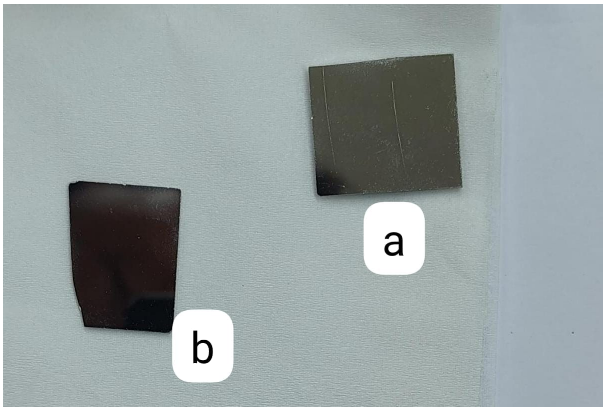

Figure 1: Aluminum thin films deposited on glass substrates.

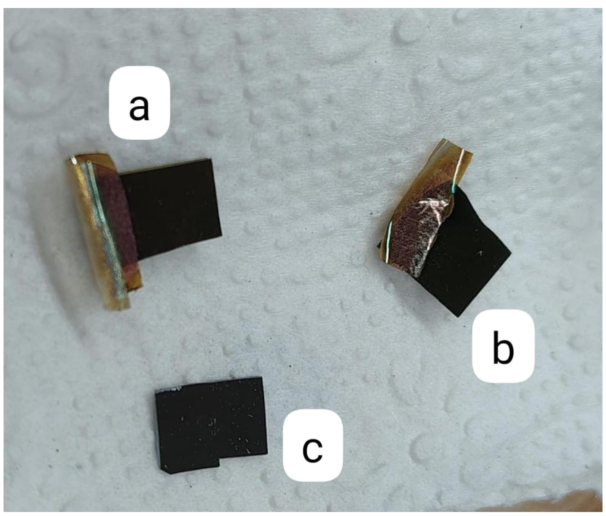

Figure 2: Silicon thin-film samples: (a,b) p-type Si thin films deposited on p-type Si substrates, removable cello tape for masking, and (c) pristine p-type Si substrate.

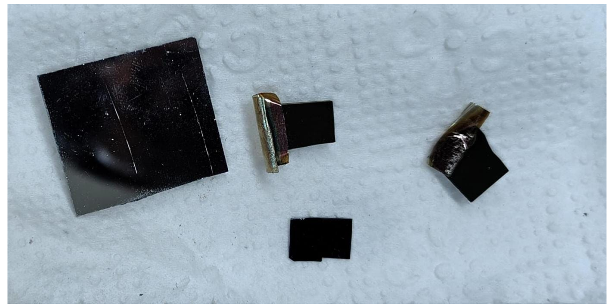

Figure 3: Overview of all prepared samples, including Al thin films on glass, p-type Si on p-type Si, and the unmodified p-type Si substrate.

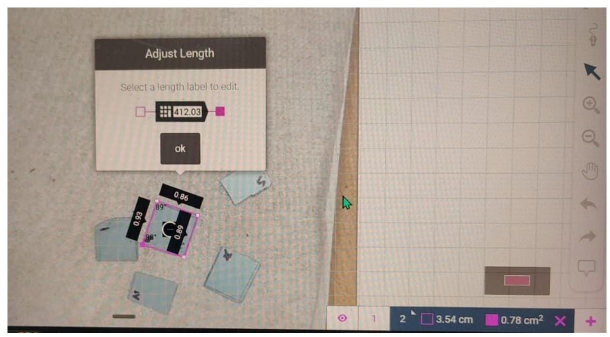

Figure 4: Surface area calculation depiction using image and calc software, by marking each substrate.

Figure 5: Ultrasonic cleaning process.

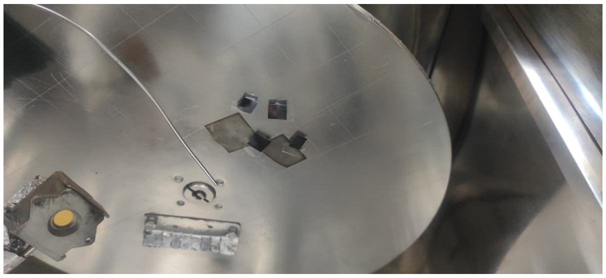

Figure 6: Substrate attaching to circular plate inside vacuum chamber.

Figure 7: substrate marked and weighted as in ref. table

Sample No.	Mass (g)	Surface Area (cm^2)
1	0.063	—
2	0.068	—
3	0.078	0.78
4	0.078	0.78
5	0.073	—
6	0.069	—